

International Islamic University Chittagong
Center for General Education
Midterm Examination, Spring-2024

Course Title: Life and Teachings of Prophet (SAAS)
Course Code: URED-3604 (URED-3201 for LLB)

Time 1:30 Hours

Full Marks: 30

[Answer the following and all questions are of equal value]

SL	Questions	Bloom's Taxonomy Domain	CLOs
1	Explain the blessings of Islam comparing it with religious and social conditions of <i>Ayyam Al-Jahiliyah</i>	C	1&2
2	Analyze the various attempts of Quraysh to stop the Prophet Muhammad (SAAS) from preaching Islam.	A	1&2
3	Give an account of the life of Prophet (SAAS) up to his prophet hood deriving some important lessons for a peaceful life. Or Write short notes along with deriving lessons: a. Migration to <i>Abisinia</i> b. <i>Isra</i> and <i>Miraj</i>	R	1&2

International Islamic University Chittagong
Center for General Education (CGED)

Midterm Examination: Spring 2024
Course Code: GEHE-3601

Program: Undergraduate
Course Title: History of Emergence of Bangladesh
Full Marks: 30

Time: 1 hours and 30 minutes.

Instructions:

- i. All Questions are Compulsory.
- ii. Figures in the right margin indicate full marks.
- iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.					
Letter of Symbol	R	U	App	An	E
Meaning	Remember	Understand	Apply	Analyze	Evaluate

	Text of the Questions	Marks	Bloom's Level	CLO
1	Sketch out the Physiographic features of Bangladesh and explain the geographical impacts on the life and society of the country.	10	An	CLO1
2	Analyze the political events and factors that led to divide India and the creation of Pakistan in 1947. Or Evaluate the movement for the 'United Independent Bengal' in 1947 and analyze the causes of its failure.	10	An	CLO2
3	"Language Movement of 1952 was a new Chapter in terms of identity of the Bengali Nationalism." Analyze the significances of the language movement and its various stages in the light of above statement.	10	E	CLO2

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Midterm Examination

Course Code: CSE 3525

Total marks: 30

Spring 2024

Course Title: Data Communication

Time: 1 hour 30 minutes

		P	C	D
1.				
a) ✓	Name the four basic network topologies and cite the advantages and disadvantages of each type.	5	1	2
b) ✓	Explain the essential components of a data communication system with figure and example.	5	1	1
2.				
a) ✓	Data communication among devices is a complex task. Explain how we simplify it.	5	2	3
b) ✓	What do you mean by transmission impairments, and how can they be mitigated?	5	2	3
✓	OR Assume that $SNR_{dB} = 36$ and the channel bandwidth is 1 MHz. What are the appropriate bit rate and signal level?	5	2	3
3.				
a) ✓	What are the advantages of optical fiber compared to twisted-pair and coaxial cables?	5	2	3
b) ✓	Carry out PCM for an sinusoidal analog signal that has highest amplitude of +16V and lowest amplitude of -16 V and assume you are using 4 bits to represent each sample.	5	2	3
✓	OR Construct digital signal with the following line coding schemes for the following bit stream: 01010000111 i. NRZ-I ii. Manchester iii. Differential Manchester iv. AMI v. AMI with HDB3	5	2	3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring 2024

Course Code: CSE 3635

Course Title: Artificial Intelligence

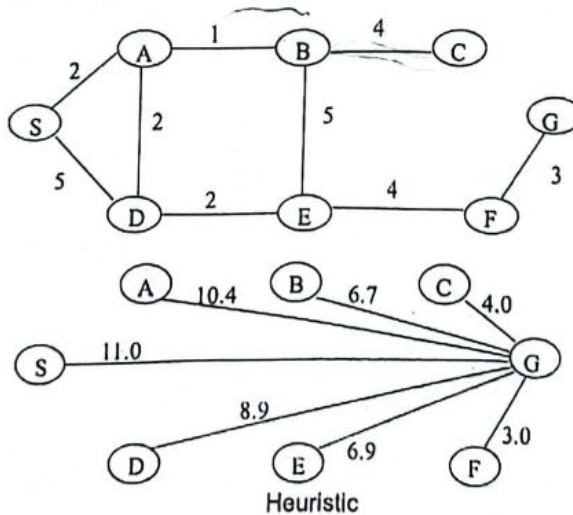
Total marks: 30 Time: 1 hour 30 minutes

[Figures in the right-hand margin indicate full marks.]

- | | Marks | CO |
|--|-------|----|
| 1. | | |
| a) "Can machine think?" –briefly elaborate the answer of this question by taking account of the "imitation game" as presented by Alan Turing. | 2 | 1 |
| OR | | |
| a) What is artificial intelligence? Enumerate the properties of an intelligent agent. | 2 | 1 |
| b) Answer the below questions considering the following agent: "Automated Taxi Driving Agent" | 2+ | 1 |
| | 3+3 | |
| i) Identify its PAGE (Percepts, Actions, Goals, and Environment) description. | | |
| ii) Characterize its environment as being either: accessible or inaccessible, Deterministic or Non-deterministic, Episodic or Non-episodic, Static or Dynamic, and discrete or continuous. You need to explain your selection. | | |
| iii) What agent architecture will be best for the Automated Taxi Driving Agent? Justify your answer with a proper illustration. | | |

2.

- a) Use the following graph to compare uniform cost search (UCS), greedy best-first search, and A* search strategies. Here, the starting node is S and the goal node is G. You need to show the expanded nodes, open list of nodes, closed list of nodes, solution path, the total number of nodes-expanded, and the cost of the solution path.



- b) How can you formulate a search problem?

OR

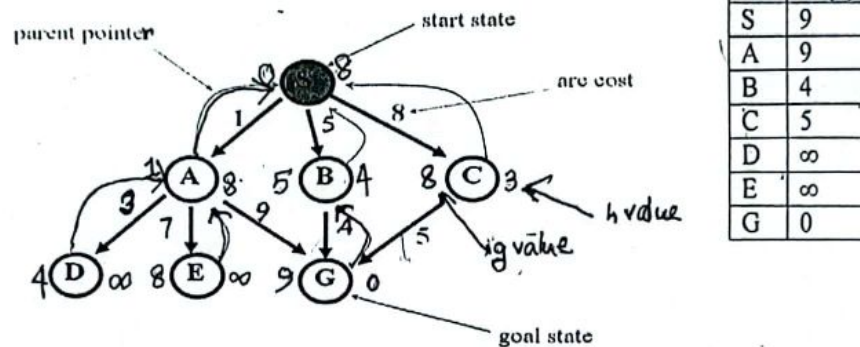
- b) "A search state keeps only the details needed for planning (abstraction)". Justify this statement with example. 2 2

3.

- a) Briefly explain the reasoning behind the following statements: 4 2
- Greedy hill climbing search has very low memory requirements.
 - The mutation step in genetic algorithm increases diversification.
- b) Write down the Hill Climbing Algorithm. What are the problems that the hill climbing search process may reach? How the problems can be dealt? 6 2

OR

- b) Suppose you have the following search space: 6 2



Assume that the initial state is S and the goal state is G. Show how A* search would create a search tree to find a path from the initial state to the goal state. If any problem arise then how can you solve it?



International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid Exam, Spring 2024

Course Code: CSE 3631

Course Title: Operating System

Time: 1 hours 30 minutes

Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1. a) Describe the interrupt handling process in an operating system. C01 03
b) How does a user program make a system call? C01 03
c) Explain the benefits of multiprogramming and multitasking. How do they improve system efficiency and user experience? C01 04
Or
List five services provided by an operating system, and explain how each creates convenience for users.
2. a) Imagine a busy internet cafe with 5 customers: C02 07
Alice (Priority 3): Needs to check her email quickly (short burst).
Bob (Priority 2): Working on a presentation (medium burst).
Charlie (Priority 1): Urgent online banking transaction (short burst).
David (Priority 3): Playing an online game (long burst).
Eve (Priority 2): Researching for a project (medium burst).
The scheduling algorithm : Preemptive Priority-based, round-robin scheduling
algorithm
Process queue: Initially, all processes are in the queue (Alice, Bob, Charlie, David, Eve).
CPU Time Slice: 2 seconds
Short Burst : 4 seconds
Medium Burst : 7 seconds
Long Burst : 10 seconds
Now draw the Gantt chart , calculate the average waiting time and CPU utilization.
- b) Compare and contrast different approaches to operating system structures: C02 03
Microkernel, Layered and Hybrid.
3. a) Compare the performance implications of context switching in different time quantum. C02 02
Or

Investigate the impact of multithreading on performance and resource utilization.

- b) Imagine a busy computer lab with four students waiting to use the printer. Schedule the students using Shortest Remaining Time First (SRTF) Scheduling Algorithm. Calculate the average waiting time.

C02 04

Student:	Arrival Time:	Burst Time (Initial):	Burst Time (Remaining):
Alice	10:00 AM	20 minutes	20 minutes
Bob	10:02 AM	10 minutes	10 minutes
Charlie	10:05 AM	30 minutes	30 minutes
David	10:10 AM	5 minutes	5 minutes

- c) Imagine a self-driving car with three critical tasks:

C02 04

Sensor Data Acquisition (Task A): Reads data from various sensors (e.g., camera, LiDAR) every 10 milliseconds (ms). Deadline: 10 ms.

Obstacle Detection (Task B): Analyzes sensor data for potential hazards every 50 ms. Deadline: 50 ms.

Path Planning (Task C): Calculates the optimal driving path based on obstacles every 100 ms. Deadline: 100 ms.

Illustrate the scheduling of these processes using Rate Monotonic Scheduling algorithm.

Or

Schedule the above scenario using Earliest Deadline First (EDF) algorithm.

//// The End////

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering

6th Semester, Mid Term Examination

Program: B.Sc in CSE
 Course Code: CSE-3637
 Full Marks: 30

Semester: Autumn 2023
 Course Title: Software Engineering
 Time: 1 hours 30 minutes

Answer the Following Three (3) Questions

				CLO	DL
1.	(a)	What is a framework? What programming languages are you familiar with?	02	CLO1	C1
	(b)	Describe the natures of software. What do you mean by "The conceptualization of the varied nature of software"?	03	CLO1	C2
	(c)	List down some segments of software cost. Do you think software maintenance is more costly than build a new software? Justify your answer.	05	CLO2	C1
2.	(a)	Describe the Software Development Process in Brief with diagram.	03	CLO2	C2
	(b)	What is "Agile" software development and what are your thoughts on it?	02	CLO3	C3
	(c)	What are your thoughts on software testing? Let's consider that you are tasked to fix a difficult bug in a large application. How do you debug the issue?	05	CLO3	C2
3.	(a)	Why do we need software requirements to develop a software? Describe about functional and non functional requirements.	4	CLO1	C4
	(b)	Briefly describe the techniques for the unambiguous specification of software. Can you describe the following meaning? function Search (X: in INTEGER_ARRAY ; Key: INTEGER) return INTEGER ; Pre: exists i in X'FIRST..X'LAST: X(i) = Key Post: X" (Search (X, Key)) = Key and X = X"	6	CLO4	C1
OR					
3.	(a)	What are the two main challenges that software firm need to face? How to overcome it?	05	CLO1	C4
	(b)	Describe about your last project you worked on, including any obstacles that you faced and your contributions to its success.	05	CLO4	C1

---THE END---

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE
Midterm Examination, Spring'24

Course Code: ECON-3501

Course Title: Principles of Economics

Time: 1 hour 30 minutes

Full Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions	
CO1	Explain the knowledge of the fundamental concepts and theories of micro and macro-economics.
CO2	Analyze the key indicators of economic growth.
CO3	Compare the economic theories and concepts to analyze behavior of individuals, firms and nations to act as a responsible citizen.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	Explain the classical definition of economics. Is economics a science or art?	CO1	U	5
1)	b)	How does production possibility curve represent the efficient use of existing technology and resources?	CO1	An	5
2)	a)	What do you mean by elasticity of demand? Explain different types of Elasticity of demand.	CO2	R	5
2)	b)	Consider the following demand and supply functions of widgets $P = 40 - Q$ (Demand), $P = 10 + 2Q$ (Supply). Find the equilibrium price and quantity. Show the equilibrium position in a graph.	CO2	Ap	5
3)	a)	What are the three basic factors of production? Describe with example.	CO3	U	5
3)	b)	Describe 'Isoquant' with an example. What are properties of an Isoquant?	CO3	E	5
OR					
3)	a)	Write short notes on: Inferior goods, Cross elasticity of demand.	CO3	U	5
3)	b)	Write short notes on: Technological impact on production, Barter transaction.	CO3	E	5

International Islamic University Chittagong

Morality Development Program (MDP)

Faculty of Science and Engineering

Mid Term Examination, Spring-2024

Semester- 6th

Course Title: Islamization of Discipline Course Code: MDP- 606

Total Marks-30

Time: 1:30 Hours

Answer the following questions, Figures in the right hand margin indicate full marks

- ✓ 1. a) Deduce the Islamic attitudes towards sciences. 2
b) What is the source of knowledge in Islamic view? 2
c) Cite about the challenges of the Holy Quran 2
d) What are the dominant elements of the universe? 2
e) In which galaxy is our solar system located? 2
- ✓ 2. a) Illustrate the Singularity and Big-Bang theories about the creation of the universe 3
b) What is Black-Hole? Mention one Quranic verse in favor of Black-Hole 2
c) "Astronomers have noticed that all the stars are moving away from each other". 5
What is the future of the universe (Open Universe or Closed Universe or Flat Universe)? Discuss from the perspective of Science and Quran.
- ✓ 3. a) What do you mean by "Nutfah" and "Sulalah"? 2
b) Describe the steps of fetus development inside the womb in the light of Quran and Science. 6
c) What is the function of Placenta? 2
- Or
- a) How to maintain good health according to Islamic approaches? 5
b) State comprehensively the parameters of self-control, exercise and relax. 3
c) Cite the Islamic View point of the environment on the perspective of the preservation of trees, woodland and green areas. 2